

Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-TERM EXAMINATION

Term 1, 2023

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	90 minutes
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 20% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 5 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**. All booklets need to be returned at the end of the exam, even if empty.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

- a. (10 marks) For the circuit shown in Figure 1,
- (5 marks) Calculate current and power of the independent voltage source.
 - (5 marks) Calculate voltage v_1 .

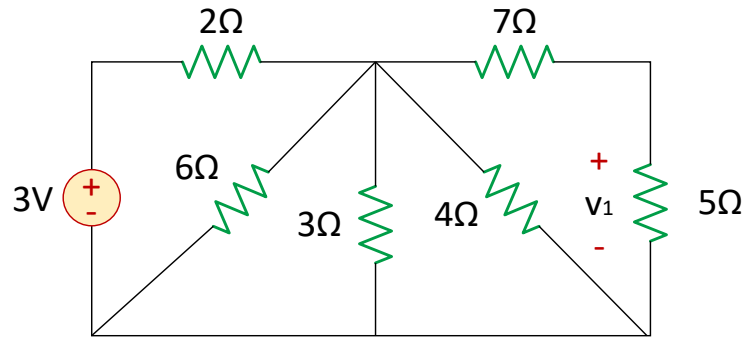


Figure 1

- b. (10 marks) The circuit in Figure 2 represents a solar-powered waste compactor. When the waste level reaches a threshold, an LED turns on and the motor starts compacting waste.
- (8 marks) What is the current through the LED?
 - (2 mark) Calculate the power dissipated by the LED protection resistor (i.e., 50Ω resistor).

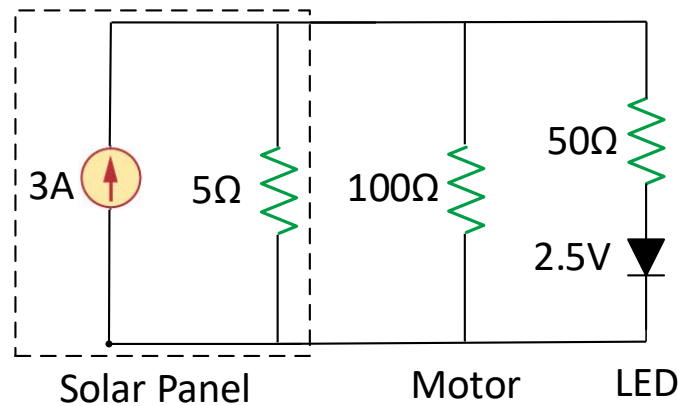


Figure 2

QUESTION 2 [24 marks]

- a. **(10 marks)** Transistors can serve as amplifiers, producing both current gain and voltage gain. For the transistor amplifier shown in Figure 3, apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a, b, c and d can be positive or negative numbers, but **DO NOT** solve them.

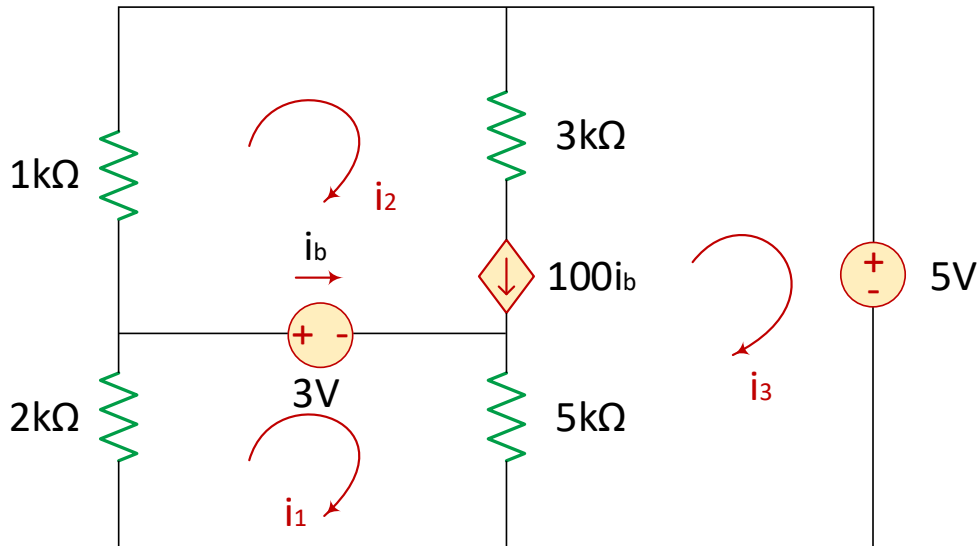


Figure 3

- b. **(14 marks)** The circuit shown in Figure 4 represents an inverting operational amplifier used to increase the level of a particular signal so that it can be transmitted more efficiently.
- (12 marks)** Use nodal analysis to calculate voltage v_o .
 - (2 marks)** Calculate gain of the circuit (i.e., the ratio v_o/v_s).

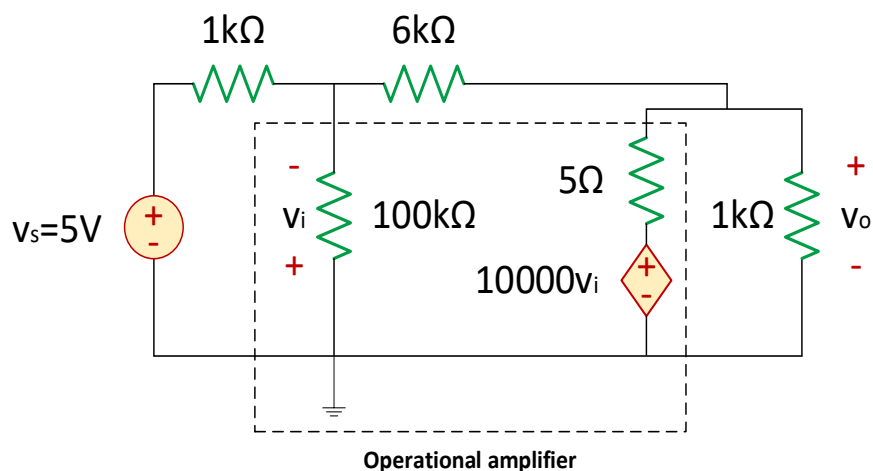


Figure 4

QUESTION 3 [24 marks]

- a. (12 marks) For the circuit in Figure 5 determine i_o using the superposition theorem.

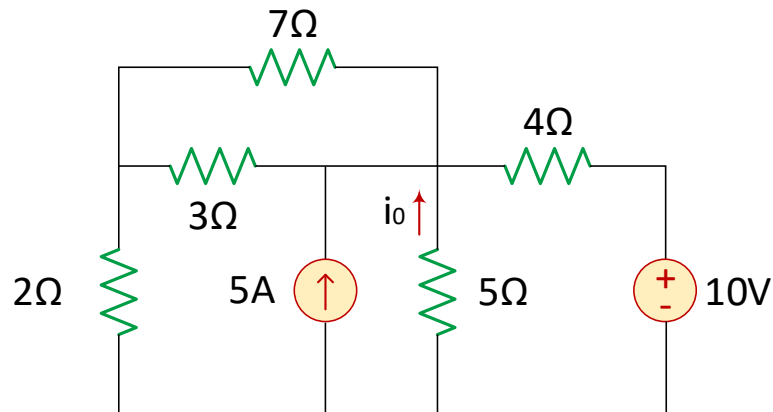


Figure 5

- b. (12 marks) For the circuit shown in Figure 6, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single voltage source and a single resistor, as seen from terminals a-b. Then, determine the Thevenin voltage, V_{Th} , and Thevenin resistance, R_{Th} , at the terminals a-b.

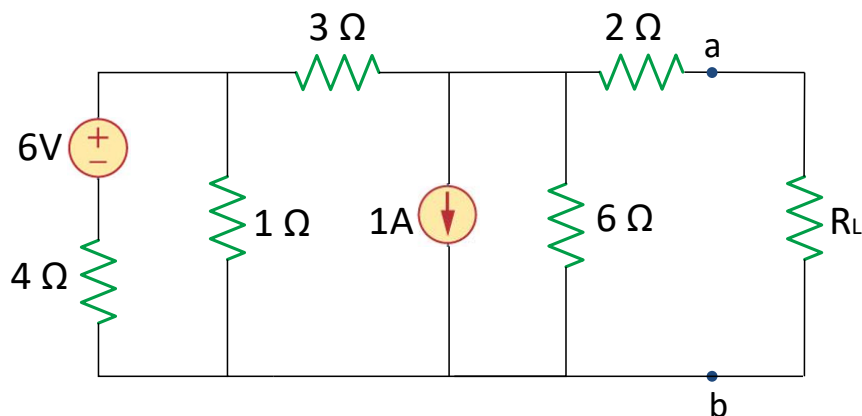


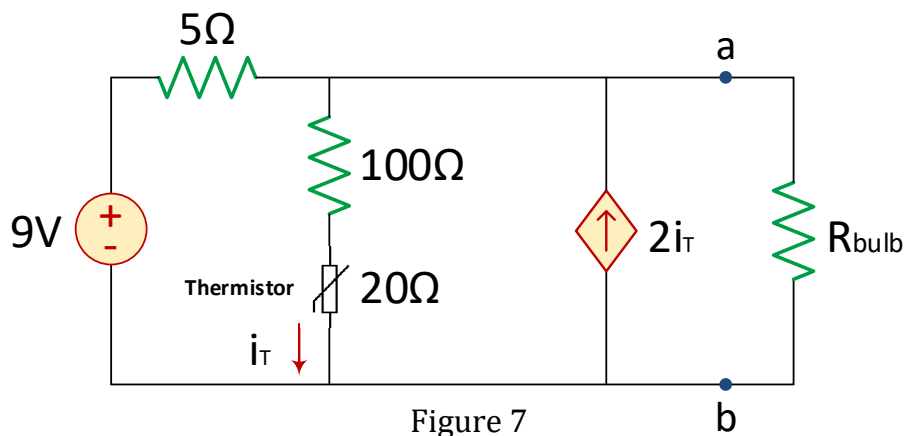
Figure 6

QUESTION 4 [14 marks]

A thermistor is an electronic component that changes its resistance based on temperature (at low temperature its resistance is large and at high temperature its resistance is small).

The circuit shown in Figure 7 is used to automatically increase the brightness of a light bulb as temperature increases. In a hot day, the resistance of the thermistor is 20Ω (as shown in Figure 7). In this situation, calculate:

- (6 marks)** Thevenin voltage seen from the bulb (i.e., from terminals a-b).
- (6 marks)** Resistance of the bulb (R_{bulb}) that will ensure the maximum power transfer from the circuit to the bulb.
- (2 marks)** Maximum power received by the bulb.



QUESTION 5 [18 marks]

There is no energy stored in the capacitor in the circuit shown in Figure 8 when switch 1 closes at $t = 0\text{s}$. Switch 2 closes 2 ms later.

- (7 marks)** Calculate $v_c(t)$ for $0 \leq t < 2\text{ ms}$.
- (11 marks)** Calculate $v_c(t)$ for $t \geq 2\text{ ms}$.

